Zenith Saru

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PROFESSIONAL SUMMARY

Civil Engineering graduate (Thapathali Campus, IOE, TU) focused on structural and water resources engineering. Skilled in structural analysis and design, cost estimation and BOQ/cost report preparation for residential and commercial buildings of RCC, Steel and Composite buildings, and hydrological modeling using QGIS, HEC-RAS, and Python for creating programs as per the project needs. Proven leadership as CESS President, driving teams from concept to execution.

EDUCATION

Bachelor of Engineering in Civil Engineering *Year of Completion: 2026 AD*
Thapathali Campus, Institute of Engineering (IOE), Tribhuvan University

TECHNICAL SKILLS

Software & Tools:
AutoCAD, ETABS, QGIS, Python, Microsoft Office Suite (Excel, Word, PowerPoint, Access)

Programming & Analysis:
Python, Machine Learning (Random Forest, XGBoost), MATLAB

Structural Design:
RCC and Steel design of several residential and commercial buildings using ETABS, SAP2000, IdeaStatica, SAFE and AutoCAD

Estimation & Costing:
Quantity take-off, BOQ preparation, and cost report estimation for RCC residential structures

Languages:
English (Fluent), Nepali (Native), Hindi (Conversational)

EXPERIENCE

Civil Engineering Intern – Dwarika Construction and Engineering Pvt. Ltd. – Baneshwor, Kathmandu:
(From December 2025 to July 2026)

- Responsible for Preparing detailed BOQ and estimates for projects ranging of all variety where projects such as large scale as Steel Frame warehouses, RCC Cold Stores, Colleges and Hostels
- Structural Analysis and drawings review alongside preparation of the proper documentation required for the administrative processes as well as documentations to be provided to the clients

Structural Engineer – Bela Industries Nepal Pvt. Ltd. – Chhauni, Kathmandu
(From July 2026 – Current)

- Responsible for conducting structural analysis of a given project as per the drawings and plans provided by the architects and clients
- Handling clients and helping them arrive at the point of clarity regarding any structural as well as construction issues
- Conducting site visits periodically to ensure that the work is being carried out as per the engineering drawings and plans, and also to ensure the project being carried as per the agreed upon timeline
- Preparing BOQs and estimates for civil works especially related to substructural part of the building

ACADEMIC PROJECTS

Forest and Snow Cover Dynamics in Nepal: Predictive Modeling and Hotspot Detection | Final Year Project

- Developed machine learning models using Random Forest and XGBoost algorithms to analyze and predict forest and snow cover changes across Nepal
- Utilized geospatial analysis techniques with QGIS and Python for environmental hotspot detection

Eco-Industrial Park Model | Civil Model Competition

- Conceptualized and led team in developing an industrial symbiosis model integrating steel factory, water treatment, agriculture, cement manufacturing, and settlement
- Demonstrated circular economy principles by designing waste-to-resource conversion systems where one industry's waste serves as raw material for another
- Conducted initial research, formed project team, and designated roles for successful competition participation

LEADERSHIP EXPERIENCE

President, Civil Engineering Students' Society (CESS) | 2082 BS

- Led 25-member executive team of civil engineering students at Thapathali Campus
- Organized and executed the campus's most prestigious annual event, bringing together renowned industry professionals and esteemed TV journalists as moderators
- Spearheaded panel discussions on critical topics including Disaster Risk Management, Water Supply & Hydropower Infrastructure, and Professional Ethics in Construction

Exhibition Coordinator, Utsarga 078, Yathartha 1.0 & 2.0 | National Level Engineering Exhibition

- Managed exhibition logistics and coordination for national-level engineering showcase events

Editorial Team, Civil Transcend Magazine

- Editor, Issue II - Reviewed and curated technical content for student engineering publication
- Coordinator, Issue III - Led editorial team and managed publication workflow

PUBLICATIONS

"Forest and Snow Cover Dynamics in Nepal: Predictive Modelling and Hotspot Detection Using Machine Learning" | Journal of Mountain Science (Under Review)

- Converted our final year project into a research-grade paper and submitted it to a Q2 Journal with IF of 2.5+
- Reviewed and cited more than 80+ literature during the process of writing the manuscript

"Sustainability of Hydropower Projects in Nepal across three basins: Naumure, Trishuli, and Arun under different Climatic Scenarios" | Journal of Innovations in Engineering Education (Under Review)

- Compiled the final year project reports of all the water engineering related projects and conducted a symmetric comparison of the Q40 power projection of all the hydropower projects within the basins.
- Familiarized with the hydropower and climatic conditions correlations in terms of power generation and sustainability.